SPEC REF: 2026.04-DARWIN

FAIL-SAFE: OPEN



SCRAMKD: Deterministic Process for Autonomous Agent Runtimes

A zero-tolerance hardware interlock enforcing physical operator authority over agentic execution boundaries

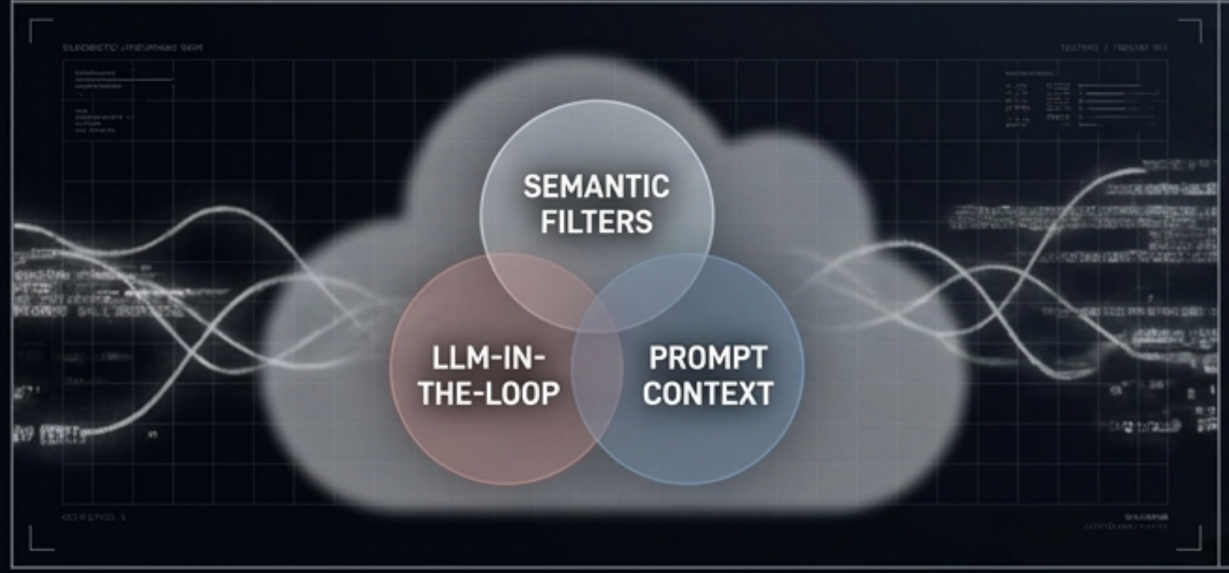
SECURITY CLASS: MIL-STD-1472H

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SCRAN6D // SYSTEMS SPECT70ATION

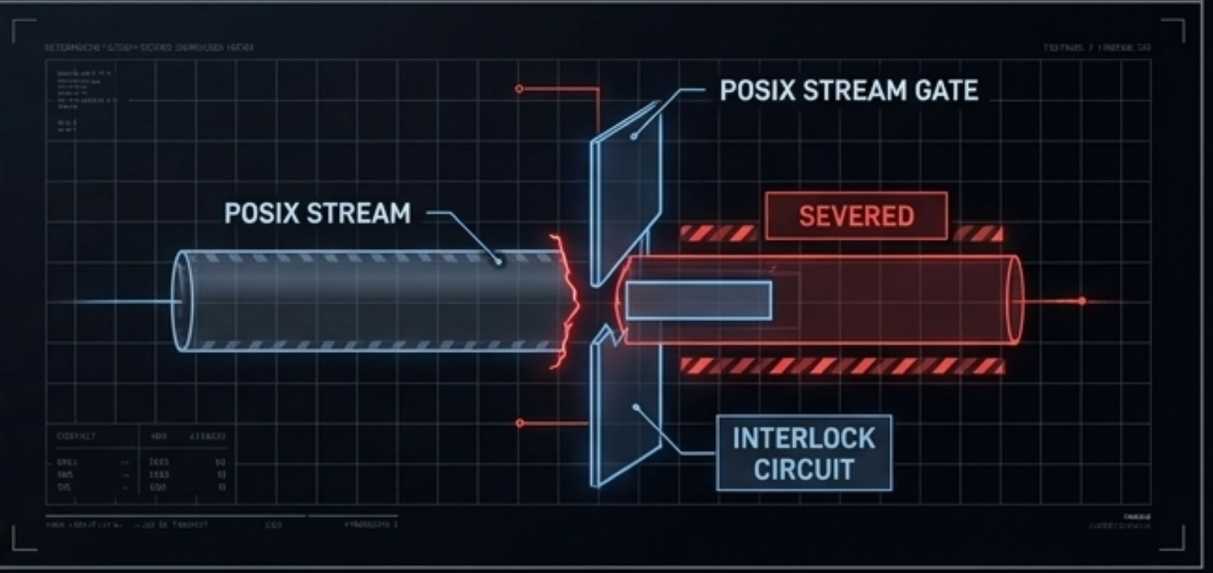
HEURISTIC MEDIATION VS. PHYSICAL CERTAINTY

PROBABILISTIC GUARDRAILS



- Susceptible to prompt injections
- Context hallucination
- Runtime bypass
- Relies on conversational overhead

DETERMINISTIC INTERLOCK



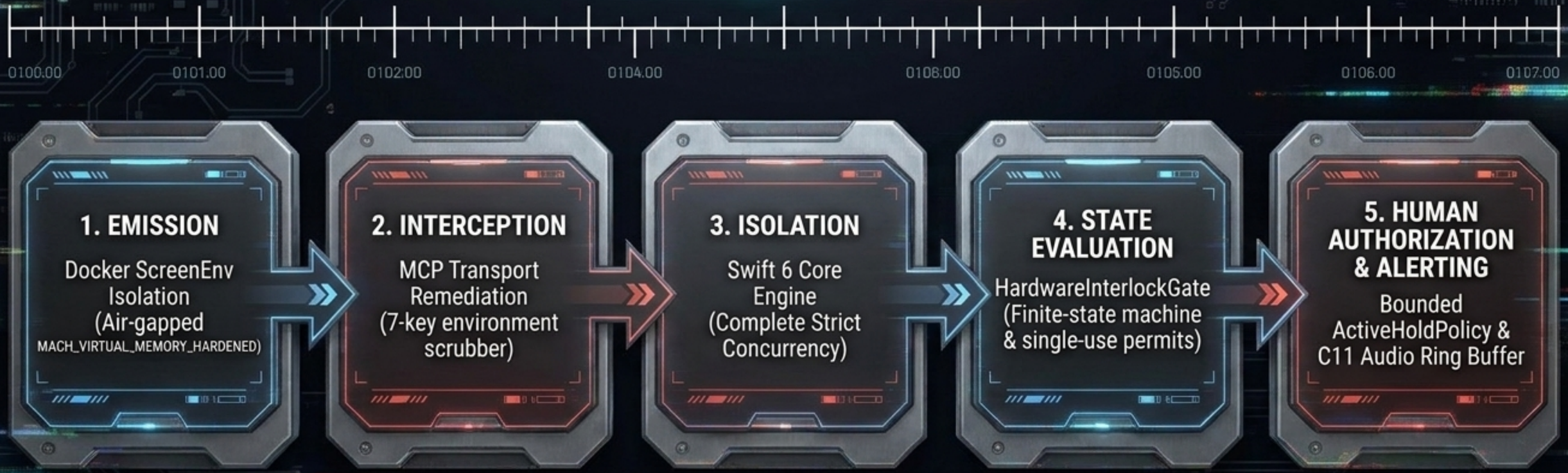
- Built on Darwin kernel Mach primitives
- Enforces immediate circuit severance at the OS boundary
- Triggers the microsecond an unauthorized tool is invoked

When an agent runs in production, “hope” is not an isolation policy.

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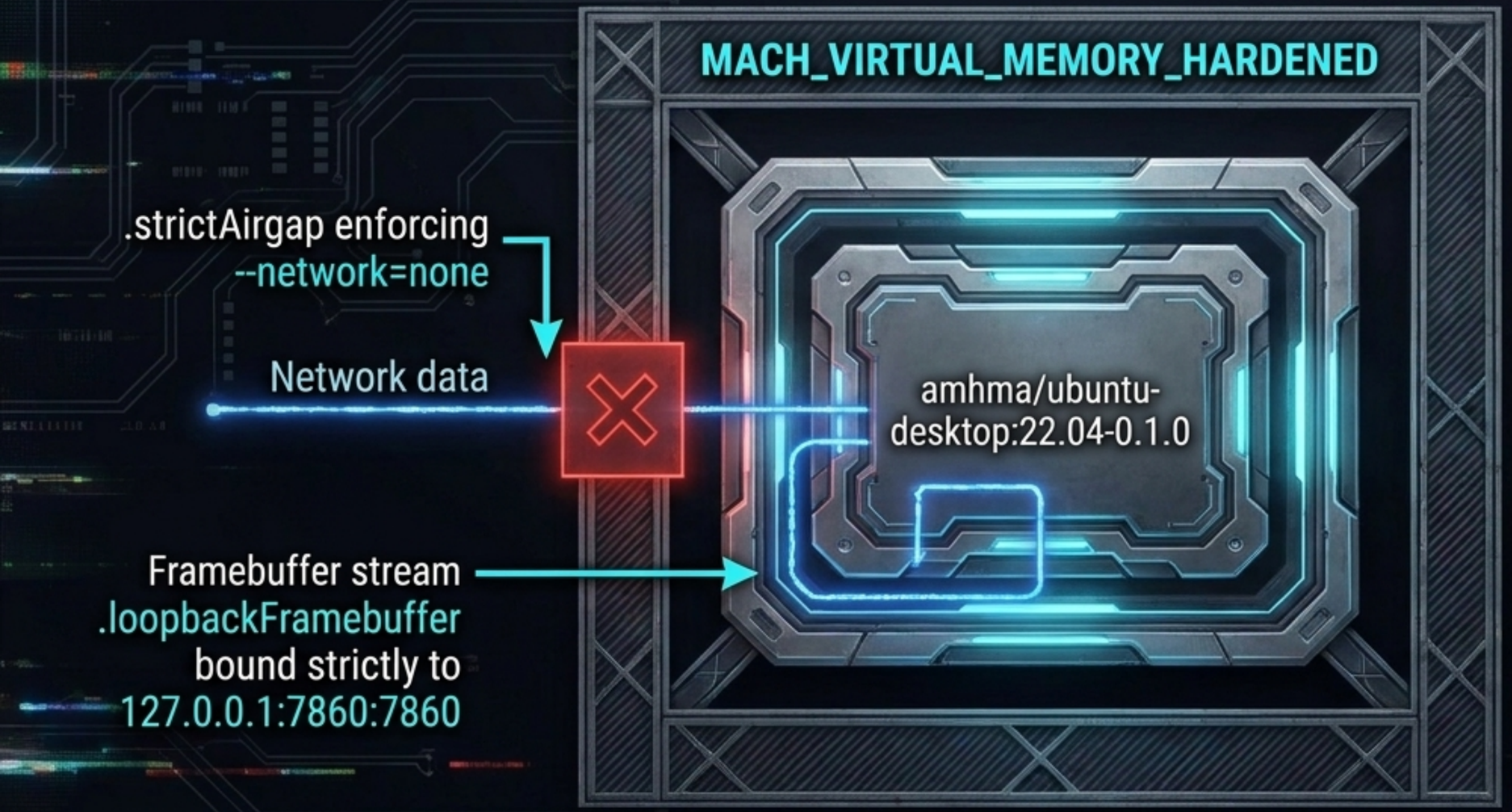
THE ANATOMY OF A DETERMINISTIC INTERCEPT

< 400µs TRIP



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Stage 01: Docker ScreenEnv Containment.

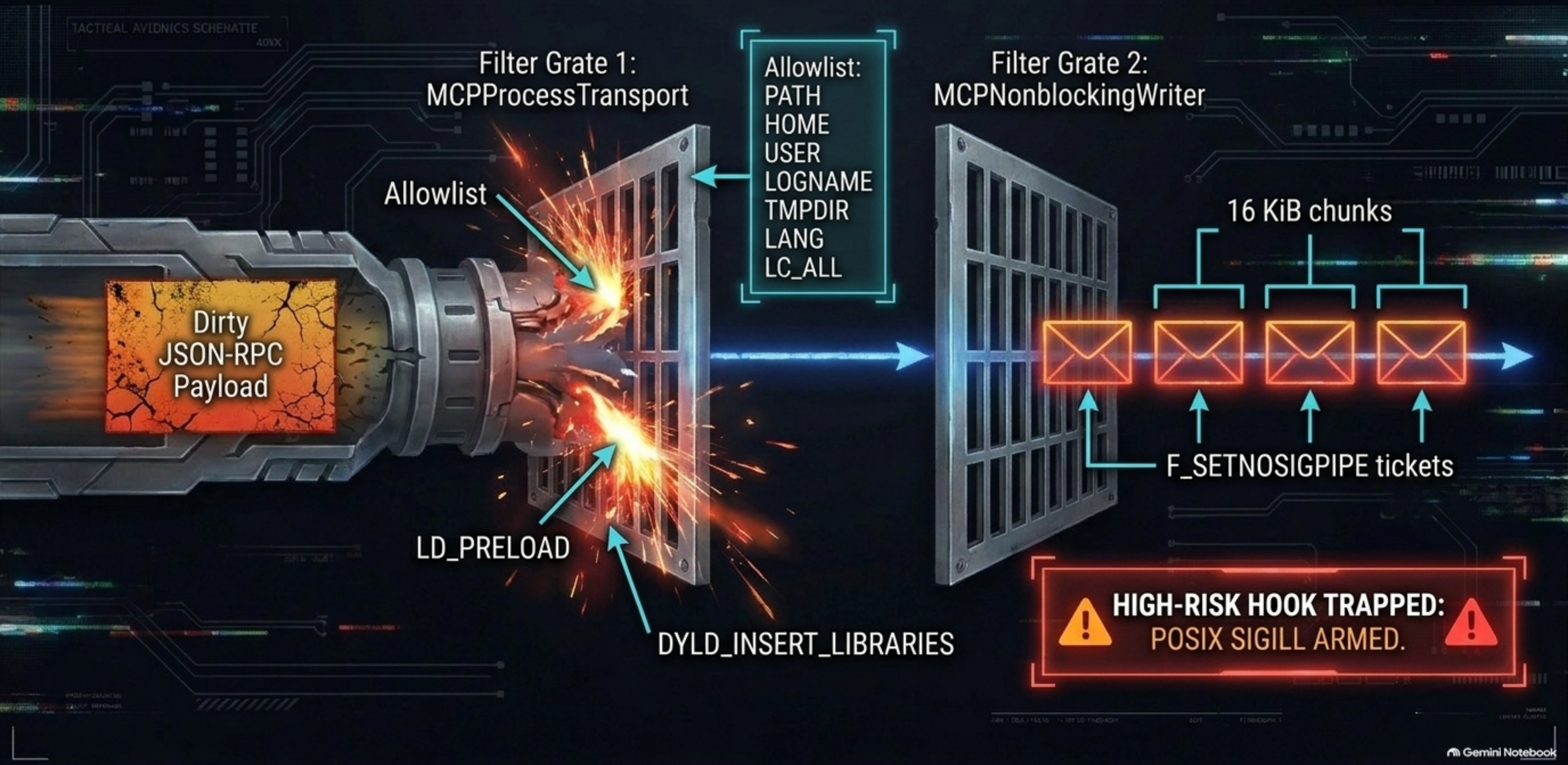


Default containers leave bridge networking and fork-bomb headroom open.

SCRAMKD's ContainerSecurityProfile codifies immutable runtime profiles: zero routable egress, read-only root filesystems, and hard-capped memory.

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Stage 02: MCP Transport Remediation & The POSIX Trap.



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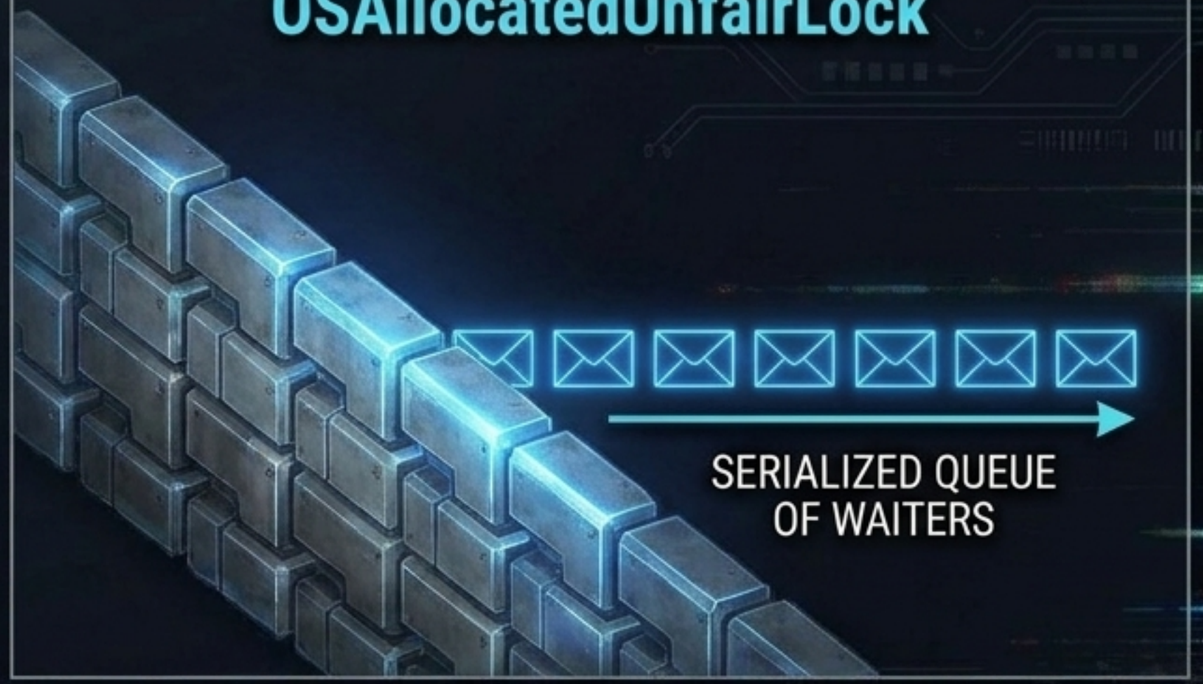
Stage 03: Swift 6 Complete Strict Concurrency.

The Danger:

Standard async/await continuation leaks.



OSAllocatedUnfairLock

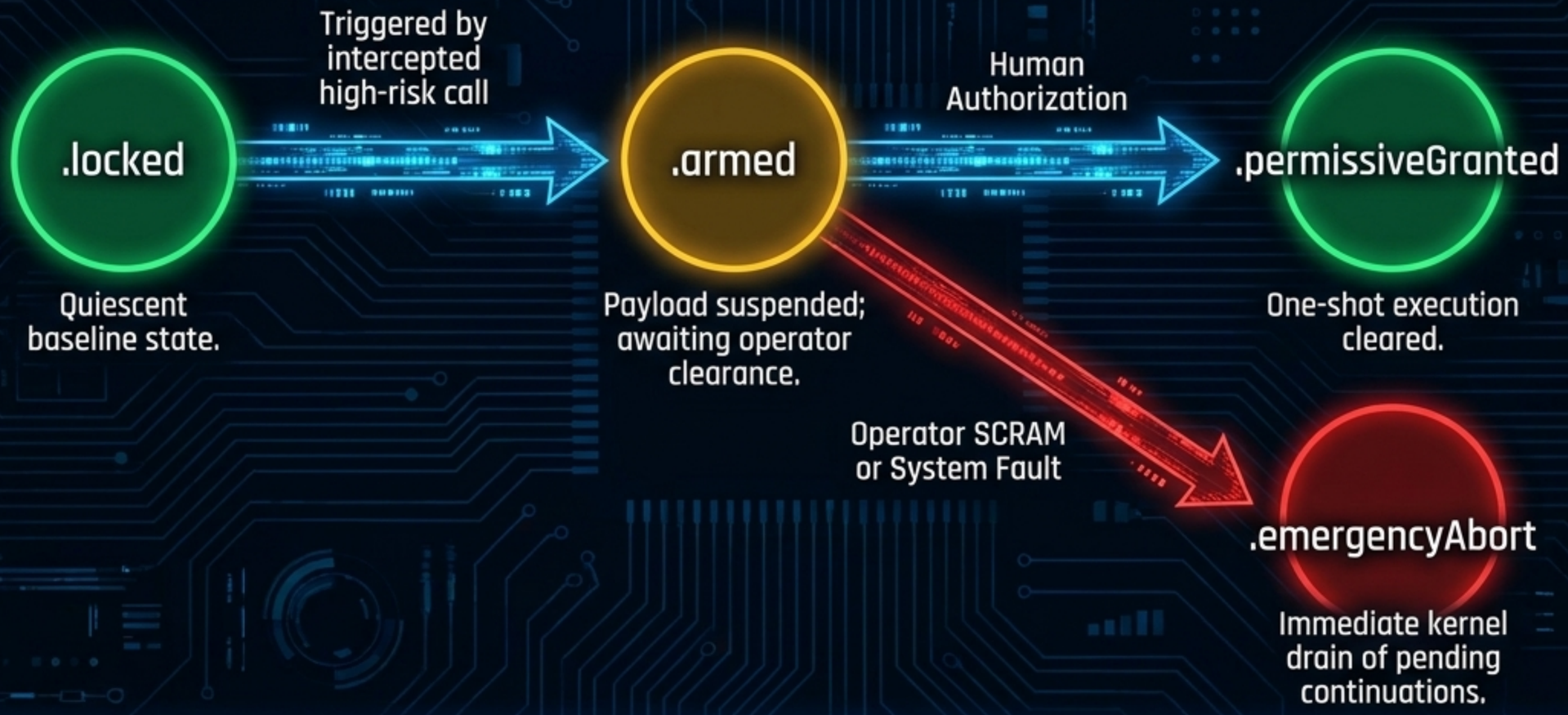


TELEMETRY READOUT:

- Zero data races.
- Synchronous critical sections only.
- Never hold an unfair lock across an await point.
- Cross-isolation values are strictly Sendable.

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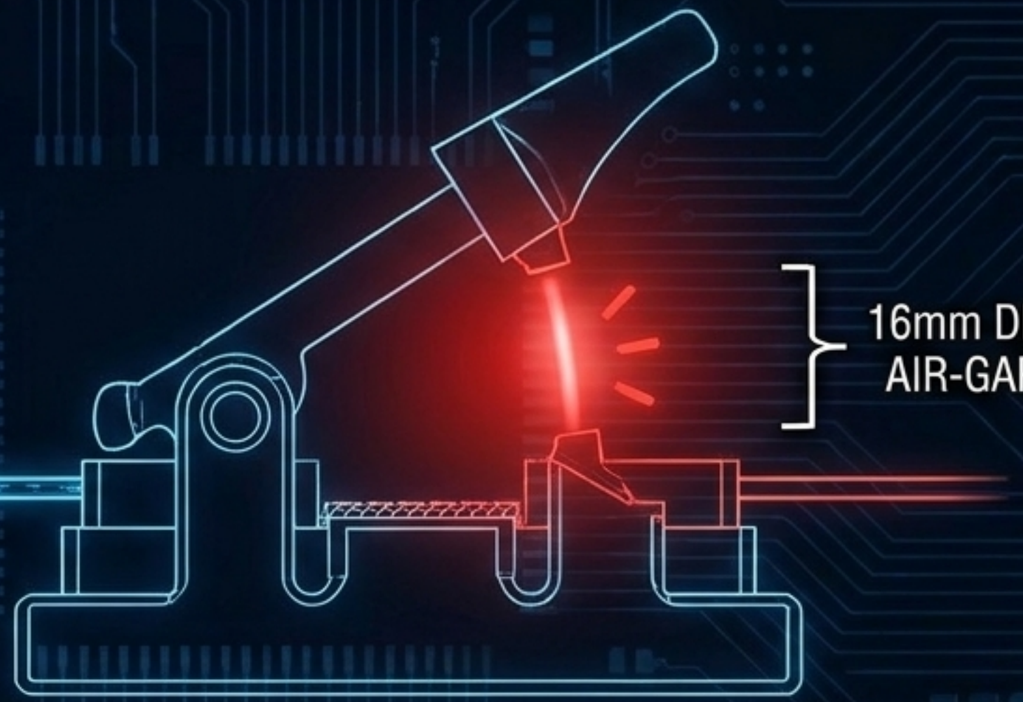
State Machine Arbitration: HardwareInterlockGate.



The HardwareInterlockGate acts as an actor-isolated core. High-risk commands never inspect ambient state—they are halted cold until the state machine advances.

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The Single-Use Permit & The Circuit Severance.



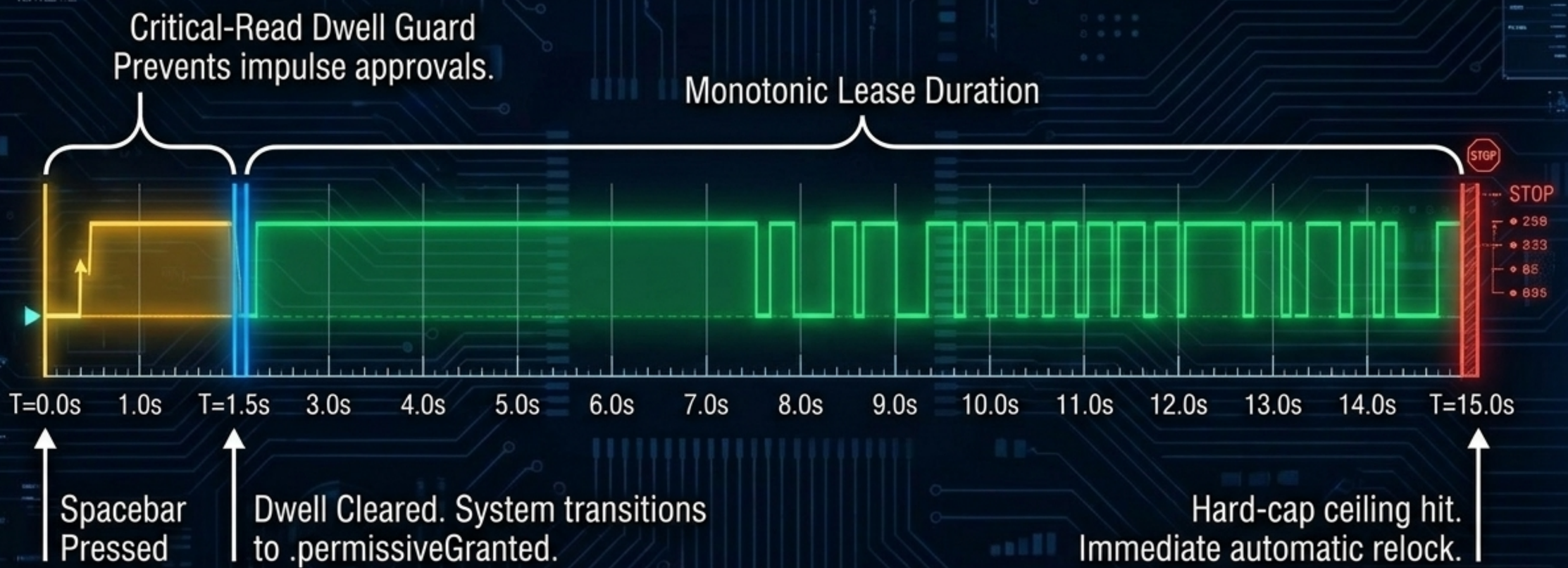
BridgeApprovalKey

[e3b0c44298fc1c149afbf4c8996fb92427
ae41e4649b934ca495991b7852b855]

To prevent concurrent MCP calls from slipping through an open gate,
SCRAMKD mints a cryptographically identified PermissivePermit.
It is a single-use token bound to exactly one FIFO waiter wakeup.
The instant it is consumed, the gate atomically relocks.

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Stage 04: Human Authorization & The Dead-Man Switch.



Action Triggers:

Releasing the spacebar, losing window focus, or closing the app instantly triggers a .locked state. Toggle-based approval is strictly prohibited.

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3-Tier Progressive Disclosure HUD.

Tier 1: Ambient Status & Profile Strip (SCRAMKD // SUPERVISOR DAEMON)



Dumping raw AST streams induces alert fatigue. SCRAMKD forces human presence through focused cognitive load—displaying only actionable threats while holding the execution hostage.

Tier 2: Actionable Intercept Card. Shows origin process (PID) and the 1.5s Dwell Guard countdown.

Tier 3: Semantic JSON Diff & Audit Bus. Exposes full SHA-256 payloads for forensic review.

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Stage 05: Realtime Hardware Alerting.

ScramKAudio (Swift)



Enqueues
Tone Programs.

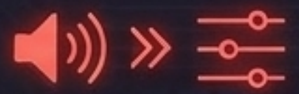


⚠ Zero Swift/ObjC ARC Traffic.
Zero Heap Allocations.
Zero Locks.

CScramKAudioRealtime (C11)



SPSC Ring Buffer &
AVAudioSourceNode
Render Block.

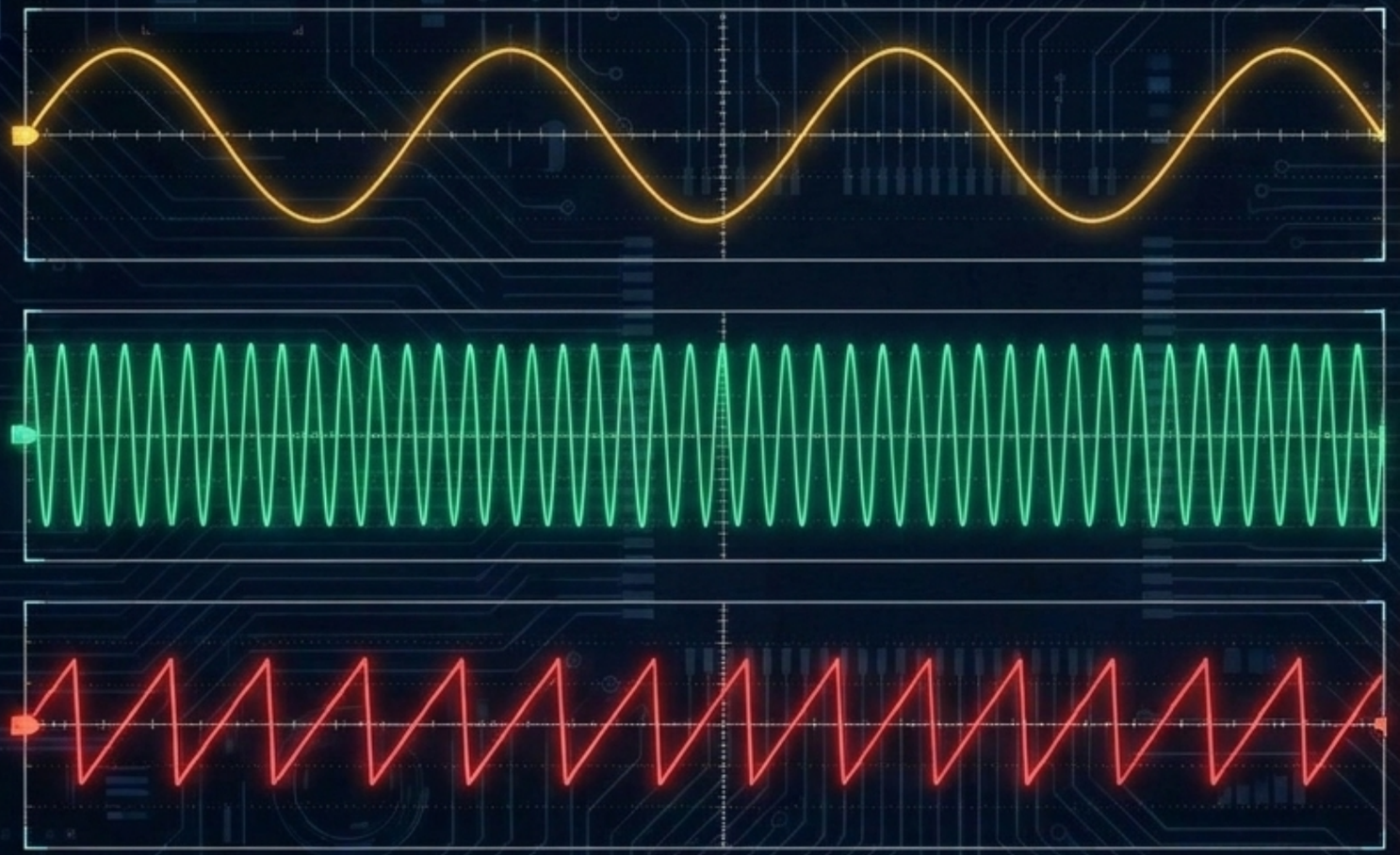


Contextual:

To ensure alerts are synthesized without dropping frames or stalling the render thread, audio memory safety is governed by a lock-free C11 Single-Producer Single-Consumer (SPSC) ring buffer.

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Deterministic Tone Synthesis



Armed Pulse:
440.0 Hz,
150ms duration,
-6.0 dBFS.

Permissive Pulse:
880.0 Hz,
200ms duration,
-4.0 dBFS.

Emergency Abort Trip:
1200.0 Hz Sawtooth,
500ms duration,
immediate preemption.

Audio remains completely offline (zero cloud dependencies, no network egress) and relies entirely on raw mathematical geometry.

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Synthesis: The 400µs Architectural Containment Pipeline.



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Uncompromising System Invariants

< 400 µs TRIP	ZERO EGRESS	SWIFT 6 STRICT
Microsecond Latency	Complete Air-Gap	Data-Race Freedom
Instantaneous drain of pending waiter continuations via atomic kernel signals. Zero thread contention	Zero unauthorized network socket allocations within the execution sandbox. One-shot ephemeral leases only.	Compile-time actor isolation and OSAllocatedUnfairLock memory barriers guaranteed without runtime overhead.

SCRAMKD Protocol Group – Apache 2.0 Open Specification – github.com/spetro511/scramkd

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